

Handling low-coverage uncertainty in methylation estimates

Abstract

Using a beta binomial/empirical Bayes estimate to shrink the value of each site towards its sample mean, using either a tuned fixed strength or one determined by the noise-adjusted variance, reduces methylation estimation error $\sim 3X$ at low coverage vs a naive estimate with no cost at high coverage. The results were determined using a simulated data set with specific numbers only being illustrative of the difference between methods tested. The algorithm is ready to be used on real data to determine its true effectiveness.

The Problem

A site read as 0.5 from 2 reads carries far less information than 0.5 from 200 reads. However, a naive estimator (m/n) treats them the same. With hundreds of thousands of sites each with variable coverage being input into a clock, unreliable estimates add noise which could ruin the accuracy of an otherwise effective clock.

Why Simulate

The test run for this report does not include an elastic-net regression to determine the age of the specimens from the methylation data. It only tests estimates of the p at a site against the true p of the site. This allows it to be tested more rigorously at the point where it acts without adding noise from an unrelated algorithm. However, the true p can only be known in a simulated data set and is not available in real data. Magnitudes are model-dependent; the true value of this report is the ranking of methods, to be checked against real data.

The simulator creates a true methylation value using the species and age of the specimen as well as the character of the site, how age-linked it is, and random noise. It uses a poisson distribution to determine the per-cell coverage and a binomial distribution to determine how many sites are methylated at a site given the true p .

The Five Methods

Each method works by a Beta-Binomial posterior mean: $\hat{p} = (m + \alpha) / (n + \alpha + \beta)$. They differ only in the value of alpha and beta. This formula adds alpha methylated reads and beta unmethylated reads before the estimate is calculated. If a site has few reads, the added reads will bias the data more than if it has many reads.

The “naive” method uses $\alpha = \beta = 0$. Without using Beta-Binomial, the value of p is simply m (methylated reads) / n (coverage)

“v1” uses $\alpha = \beta = 1$. This shrinks every site towards 0.5 with a low strength.

“v2” uses $\alpha = \text{mean} * \text{strength}$, $\beta = (1 - \text{mean}) * \text{strength}$. This shrinks every site towards its own mean with a set strength. The value of the fixed strength can be tuned from the data to achieve the best result (a strength of 20 was used for this report).

“v3” tunes the strength automatically and individually from each site's variance. Therefore, the more “spread out” a site's reads are across samples, the less they will be pulled towards the center and vice versa.

“v3b” does the same as v3, however, it subtracts the sampling-noise from the variance to only consider the biological variance.

Results

Estimation accuracy vs. coverage

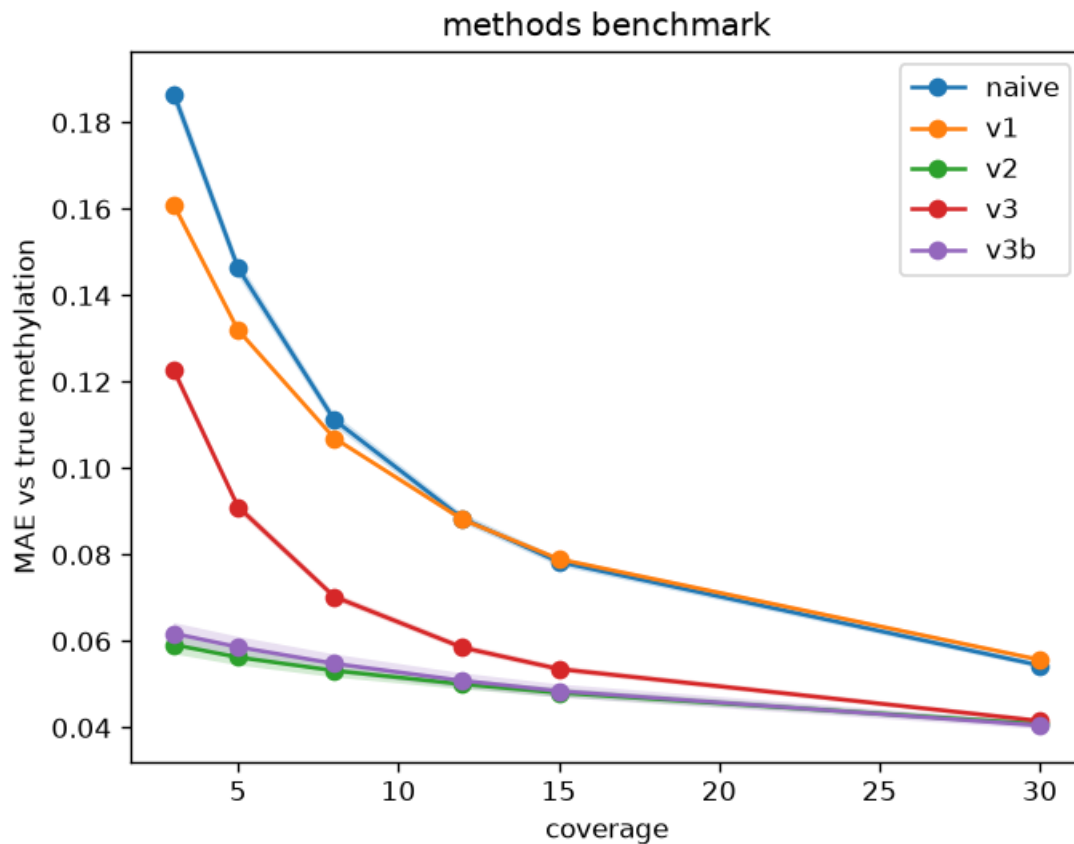


Figure 1: Mean absolute error between estimate and true methylation, vs. coverage. Shaded = ± 1 SD.

Methods v2 and v3b are the clear winners and are nearly identical. This makes sense because the only difference between them is that v3b tunes its own strength rather than needing to be tuned manually. v1 only slightly beats out the naive estimate at low coverage. v3 underperforms compared to v2 and v3b due to its inefficiency in selecting its strength from sampling variance. Methods tend to converge at higher coverage. However, the best methods still outperform the naive estimate with high coverage because they also correct some variance not caused by low coverage by biasing towards the mean.

Does a better estimate make a better clock

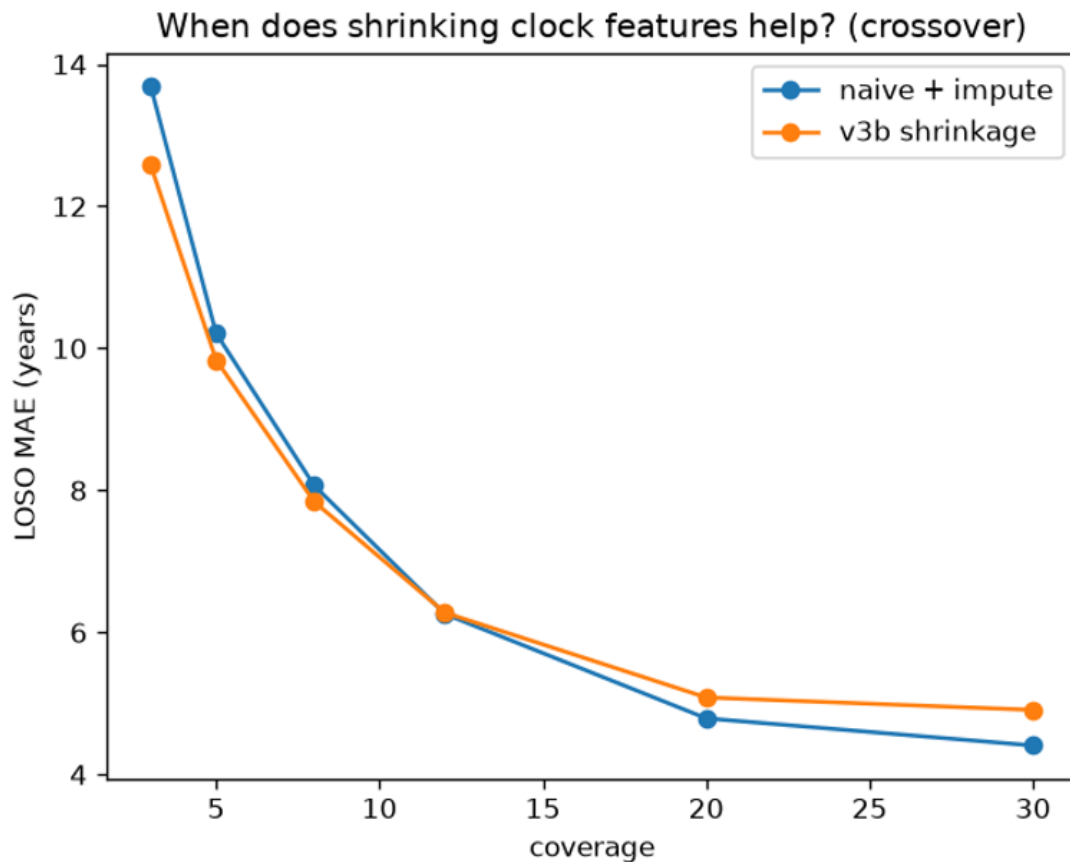


Figure 2: Leave-one-species-out clock error years vs. coverage, using shrinkage vs. naive + impute as features. Run on 3 seeds with no error bars; trend is visible, but numbers are inaccurate.

Shrinkage helps the clock mainly at low coverage. At higher coverage it performs worse than simply using m/n and imputing the zero-coverage holes. The clock already throws out most of the sites where most of the value came from in the previous test. All the shrinkage does is reduce the variation in the age-linked sites, which is exactly what we want to measure. A better estimate does not necessarily mean a better clock, it depends on the coverage.

Recommendation

- For estimating accurate per-site levels: shrink towards the site mean using v3b (parameter-free), or v2 if you tune the strength.
- For clock features, use shrinkage only when the coverage is low.
- Avoid v1 and v3, they carry no advantage.

Caveats and Next Steps

- Magnitudes are simulator specific. The first step on real data is to reproduce this logic and check the real coverage distribution.
- This simulator does not have high coverage variability within tests to make the differences between coverages easier to see. Real coverage varies far more and is worth testing.
- The clock comparison used fewer seeds and was not tested as rigorously as it was not the focus of this report. Testing these methods using a more advanced clock with more iterations is worthwhile to determine the true relationship.